

Very similar work: geometric trajectory generating

- Modeling the geometric coordinates given node features and edge set

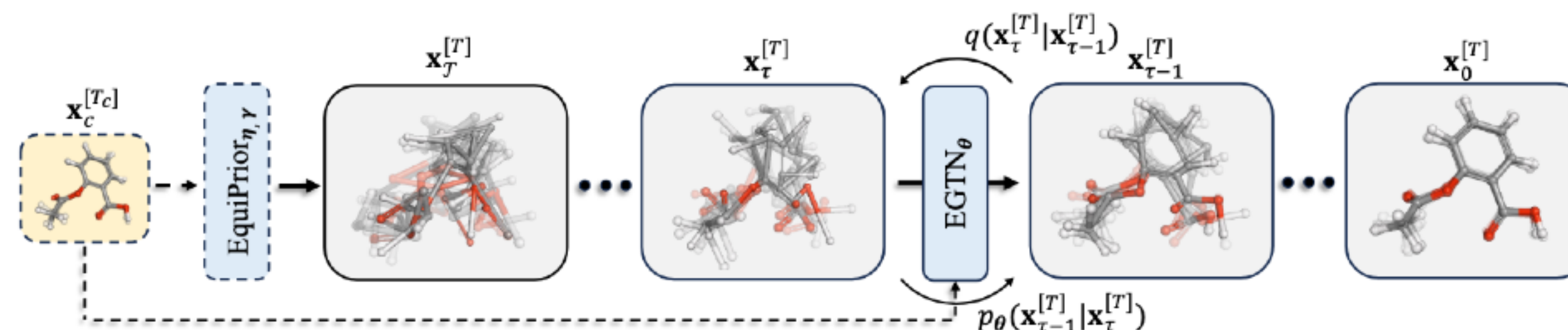
$p(\mathbf{x}^{[0,1,\dots]}, \mathbf{h}, \mathbb{E})$ (**But equivariance required**) (Han et al. 2024 neurips)

$\mathbf{x}$: node 3D coordinates

- Policy: Equivariant GNN + temporal-level feature attention

- If we view the coordinates as data on nodes, this is like our setting

$\mathbb{E}$: edge set, molecule topology



Diffusion model fashion

- If prior is SE3-inv, transition kernel is SE3-eqv., then marginals are SE3-inv

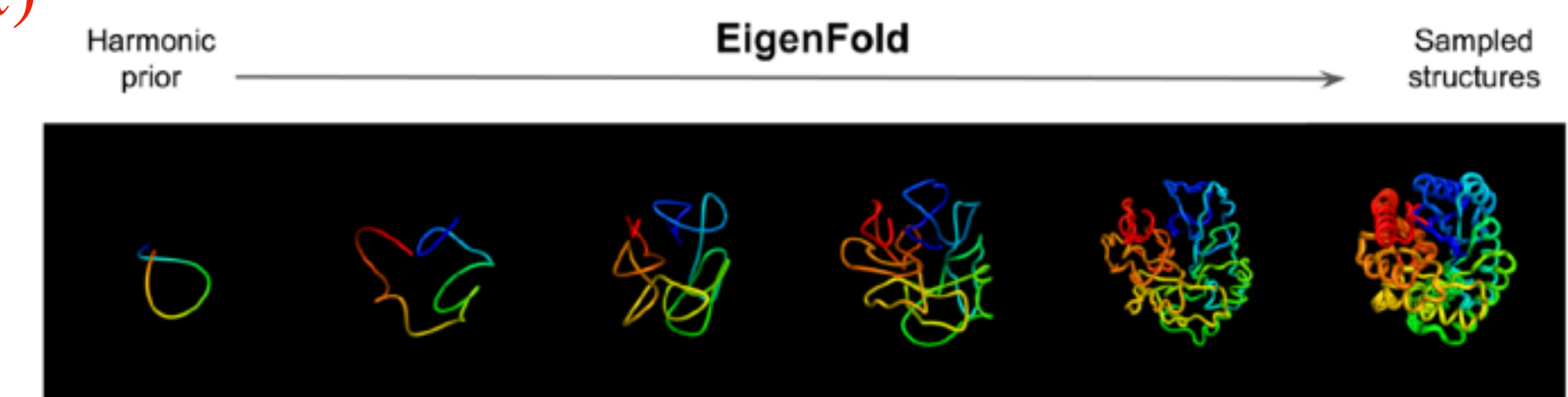
- Another Referred as Learning graph-dependent probability distribution $p_G(\mathbf{x})$

- Jing et al. 2023. Eigenfold: generative protein structure prediction with diffusion models.

- Reference process: $dx = -Hx dt + dw$, where $-Hx = \nabla E = \nabla(x^T Hx)$

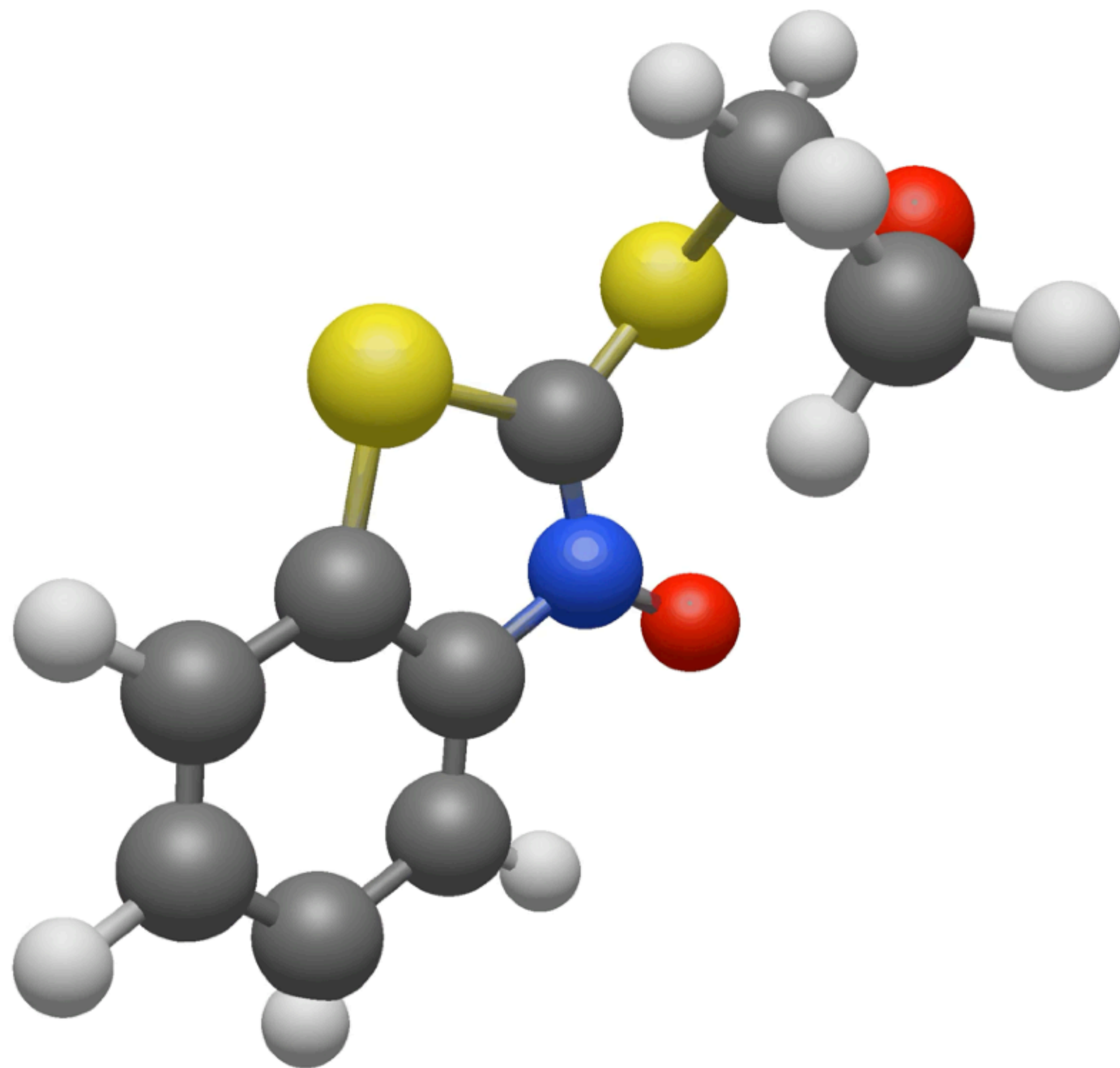
- prior: $p(x_T) \propto \exp(-0.5x^T Hx) = \exp(-0.5 \sum_{i,j \in \mathbb{E}} |x_i - x_j|^2)$

Graph GP



Similar work 2 for conformer generation

[Liu et al. 2025. Adjoint SB Sampler. Neurips] [Haven et al. Adjoint sampling. ICML 2025]



Here:

$$\exp(-\frac{1}{2}\|x_1 - x_2\|^2) = \exp(x^\top R x), \quad \text{where } R = \begin{bmatrix} 1 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & 0 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 1 \end{bmatrix}$$

- Cartisian gen: $t = 0$ (harmonic prior), e.g., graph GP: 2-particle in 3D, DW-4, LJ-15/55
- torsional generation: (tori space)
- EGNN / tensor field network policy
- directly from energy without data (SOC)**

Our work:

- more general topology distributions**
- Topological stochastic heat diffusion
$$dY_t = -cLY_t dt + g_t dW_t$$
- Topology-aware transition kernels